Algorithm 2: Function COMPUTE_CHANGESET_R for the Asynchronous Architecture

Input: RDB2RDF view $W = (V, S, M)$; update $u = (D, I)$ over relation R.
Output: Insertions on tables *rdf_maintenance_queue* and *rdf_rule_contribution*.

1. `// Event-level queue initialization:`
2. Insert one row into *rdf_maintenance_queue* for the statement-level update $u = (D, I)$;
3. `// Rule-level computation and publication:`
4. **foreach** $\Psi \in Relev(R)$ **do**
5. `// Execute the following two steps sequentially:`
6. `// STEP 1 - Incremental Maintenance Computation for` Ψ
7. $\Delta^{+}_{pivot}[\Psi](u) \leftarrow \emptyset$;
8. $\Delta^{+}_{rel}[\Psi](u) \leftarrow \emptyset$;
9. **if** Ψ *is pivot-relevant to updates on* R **then**
10. Compute $\Delta^{+}_{pivot}[\Psi](u)$;
11. **end**
12. **if** Ψ *is relation-relevant to updates on* R **then**
13. Compute $A^{-}[\Psi](u)$, $A^{+}[\Psi](u)$, $S_2[\Psi](u)$, and $\Delta^{+}_{rel}[\Psi](u)$;
14. **end**
15. $\Delta^{+}[\Psi](u) \leftarrow \Delta^{+}_{pivot}[\Psi](u) \cup \Delta^{+}_{rel}[\Psi](u)$;
16. `// STEP 2 - Rule Contribution Publication`
17. Insert one row into *rdf_rule_contribution* for Ψ with $A^{-}[\Psi](u)$, $S_2[\Psi](u)$, and $\Delta^{+}[\Psi](u)$
18. linked to the *event_id* created in *rdf_maintenance_queue*;
19. **end**